

01MAT101121903-B

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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Regular and Supplementary Examination December 2020

(2019 Scheme)

Course Code: MAT101

Course Name: LINEAR ALGEBRA AND CALCULUS

(2019 Scheme)

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

- 1 Find the rank of the matrix $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 2 & 4 & 1 \\ 5 & 6 & 7 & 5 \end{bmatrix}$ (3)
- 2 What type of conic section the following quadratic form represent?
 $Q = 17x_1^2 - 30x_1x_2 + 17x_2^2 = 128$ (3)
- 3 If $U = \frac{x^3 + y^3}{x - y}$, find $\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y}$. (3)
- 4 If $z = x^2y$; $x = t^2$, $y = t^3$ find $\frac{dz}{dt}$ using chain rule. (3)
- 5 Evaluate $\int_0^3 \int_0^2 \int_0^1 xyz \, dx dy dz$ (3)
- 6 Use double integrals to find the volume of the solid enclosed below the plane
 $z = 4 - x - y$ and above the rectangle $R = \{(x, y); 0 \leq x \leq 1, 0 \leq y \leq 2\}$. (3)
- 7 Does the series $\sum_{k=1}^{\infty} \left(\frac{4}{5}\right)^k$ converge? If so, find the sum. (3)
- 8 Test the convergence of the series $\sum_{k=1}^{\infty} \frac{k^2}{2k^2 - 1}$ (3)
- 9 Find the binomial series for $f(x) = \frac{1}{\sqrt{1+x}}$ up to third degree term. (3)
- 10 Find the Maclaurin's series for $f(x) = x \cos x$ up to third degree term. (3)

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PART B

Answer one full question from each module, each question carries 14 marks

Module-I

- 11 a) Using Gauss elimination method find the solution of the system
 $x + y - z = 9, 8y + 6z = -6, -2x + 4y - 6z = 40$ (7)

- b) Find the matrix of transformation that diagonalize the matrix

$$\begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$$
. Also, find the diagonal matrix. (7)

- 12 a) Find the value of λ and μ for which the system of equations
 $2x + 3y + 5z = 9, 7x + 3y - 2z = 8, 2x + 3y + \lambda z = \mu$ (7)

has (a) no solution (b) unique solution (c) more than one solution

- b) Find the eigen values and eigen vectors for the matrix $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ (7)

Module-II

- 13 a) If $w = \sqrt{x^2 + y^2 + z^2}$, $x = \cos \theta$, $y = \sin \theta$, $z = \tan \theta$,
find $\frac{dw}{d\theta}$ at $\theta = \frac{\pi}{4}$ (7)

- b) Find the local linear approximation L of $f(x, y, z) = xyz$ at the point
 $P(1, 2, 3)$. Compute the error in approximation f by L at the point
 $Q(1.001, 2.002, 3.003)$. (7)

- 14 a) Locate all relative extrema of $f(x, y) = x^3 y^2 (12 - x - y)$ (7)

- b) Let f be a differentiable function of three variables and suppose that
 $w = f(x - y, y - z, z - x)$, show that $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$ (7)

Module-III

- 15 a) Find the area bounded by the x -axis, $y = 2x$, $x + y = 1$. (7)

- b) Change the order of integration and hence evaluate $\int_0^1 \int_{x^2}^{2-x} dy dx$ (7)

- 16 a) Find the volume bounded by the cylinder $x^2 + y^2 = 9$ and the planes
 $y + z = 3$ and $z = 0$ (7)

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- b) Find the mass and centre of gravity of the lamina in the first quadrant bounded by the circle $x^2 + y^2 = 1$ and the coordinate planes with density function xy . (7)

Module-IV

- 17 a) Test the convergence $i) \sum_{k=1}^{\infty} \frac{k(k-1)}{(k+1)(k+2)(k+3)}$ $ii) \sum_{k=1}^{\infty} \left(\frac{k+2}{2k-1} \right)^k$ (7)
- b) Test whether the following series is absolutely convergent or conditionally convergent. $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k(k+1)}}$ (7)
- 18 a) Test the convergence of the series $1 + \frac{1.3}{1.2} + \frac{1.3.5}{1.2.3} + \frac{1.3.5.7}{1.2.3.4} + \dots$ (7)
- b) Test the convergence: $(i) \sum_{k=1}^{\infty} \frac{2}{3^k + 5}$ $(ii) \sum_{k=1}^{\infty} (-1)^{k+1} \left(\frac{k}{2k+3} \right)$ (7)

Module-V

- 19 a) Find the Fourier series of $f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi < x < 0 \\ 1 - \frac{2x}{\pi}, & 0 < x < \pi \end{cases}$ (7)
- b) Obtain Fourier series of e^x the interval $(-1, 1)$ (7)
- 20 a) Find the Fourier series $f(x) = x^2 - 2$ in the interval $(-2, 2)$ (7)
- b) Find the Fourier cosine series of $f(x) = x^2$ in $0 < x < \pi$. (7)
